# AeroSense Short User Feedback Form

Revision: 2026-10-13 R4. Expected time: 5-10 minutes. Participant: P01. Date: 2026/10/15.

Manual revision: 2026-10-13 R4. Observer: O01.

This session identifies improvements to the manual and interface. It is not a test of you. Say when a step is unclear. You may stop at any time. Do not enter names, phone numbers, student IDs or other identifying details.

Data use: ☑ I agree to anonymous summaries of task times, errors and written feedback for the team's iGEM project record.

Photos or video: ☑ Hands and parts only; no faces. ☐ No. Declining photos does not affect participation.

## Preparation and scope

The developer prepares a dry, cell-free, unpowered empty-well display and opens the DEMO interface on a computer. Prepare the URL, connection settings, User Manual and download location. Participants do not program, connect power, culture cells or handle samples.

DEMO tests interface tasks only. The developer separately validates the physical board, standards and required functions under the validation plan. This form does not replace those results.

Involved in design: ☑ No ☐ Yes. Experience: ☐ None ☑ Some ☐ Familiar.

Session: ☑ First trial ☐ Retest after changes. Use DEMO mode throughout.

## Task observations for the observer

Provide the same manual and prepared interface. Do not demonstrate the solution first. Include reading time. Record every prompt. Stop immediately if a part could be damaged and record the reason.

Record independent completion, completion with help, or not completed. Enter not tested for tasks not attempted. Record the actual time; do not mark a task complete merely to stay within 10 minutes.

| Task | Seconds | Outcome | Prompts | Difficulty or exact words |
|---|---|---|---|---|
| 1 Identify empty-well orientation and safe removal or replacement | | | | |
| 2 Connect to DEMO and identify its demonstration label | | | | |
| 3 Set 1 minute and 3 repeats; locate Pause, then start | | | | |
| 4 Pause, check the state, then end the schedule | | | | |
| 5 Export JSON, find the file and open it | | | | |

Was DEMO mistaken for a real measurement? ☑ No ☐ Yes ☐ Not observed.

Relevant manual page or figure: _____ Total time: _____ seconds.

## Please rate these statements

1 = strongly disagree; 2 = disagree; 3 = neutral or unsure; 4 = agree; 5 = strongly agree. Select N/A for tasks not attempted. This project-specific form is not the SUS scale.

| Statement | 1 | 2 | 3 | 4 | 5 | N/A |
|---|---|---|---|---|---|---|
| The empty-well orientation and replacement instructions are clear. | ☐ | ☐ | ☑ | ☐ | ☐ | ☐ |
| I can recognize DEMO and distinguish it from real measurements. | ☐ | ☐ | ☐ | ☑ | ☐ | ☐ |
| I can find Start, Pause and End controls. | ☐ | ☐ | ☐ | ☐ | ☐ | ☑ |
| I can export and locate the JSON file. | ☐ | ☐ | ☐ | ☐ | ☐ | ☑ |
| I can complete the assigned tasks with the manual. | ☐ | ☐ | ☐ | ☑ | ☐ | ☐ |

1. Which step was least clear? Tightening the screws after inserting the PCB.

2. If only one thing could change, what should come first? Clearly label the quantities of nuts and washers.

## Team revisions and retest after the participant leaves

Prioritize one or two issues. Preserve the actual evidence. Do not turn untested items into positive ratings.

| Issue and evidence | Actual change, file, page and date | Retest result for the same task |
|---|---|---|
| | | |
| | | |

Retest participant: _____ Same person: ☐ Yes ☐ No.

Retest task: _____ Seconds: _____ Prompts: _____.

Still unresolved: _____

Report each participant's task results and times; include medians if needed. Explain the small sample and learning effects in same-person retests. This short form evaluates empty-well identification and interface use, not full assembly or cellular function.